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Petrology of Apollo 14 high-alumina basalt

IKUO KUSHIRO
Geophysical Laboratory, Carnegie Institution of Washington, Washington, D.C. 20008

YUKIO IKEDA AND YASUO NAKAMURA*
Geological Institute, University of Tokyo, Tokyo, Japan

Abstract—Melting experiments and microprobe analyses have been carried out on the Apollo 14 high-alumina basalt 14310 (Al_2O_3 21.7 wt% with MgO/FeO ratio higher than those of the Apollo 11 and 12 crystalline rocks). Another crystalline rock, 14053, and a breccia, 14321, have been analyzed with the microprobe. The liquidus phase of the rock 14310 is plagioclase up to about 10 kb, chromian spinel between about 10 and 20 kb, and pyrope-rich garnet above 20 kb. The composition of this rock, as well as those of feldspathic basalts that are considered to be highland materials, is separated from possible lunar "mantle" compositions by low-temperature cotectic boundaries at least up to 10 kb, suggesting that the magmas of these rocks are not products of direct partial melting of the lunar "mantle" or of fractional crystallization of a primary magma at least to the depth of 300 km under anhydrous conditions. It is most probable that these highland rocks were generated by partial or bulk melting of a plagioclase-cumulate rock that had been formed by a large-scale differentiation in the shallow parts of the lunar interior at an earlier stage. This argument is valid even if rock 14310 is a plagioclase-cumulate rock with less than 15% plagioclase enrichment. If considerable amounts (70% or more) of alkalis were lost before solidification, the original magma could have been formed by direct partial melting of the lunar "mantle" material with MgO/FeO ratio similar to that of the earth's upper mantle or by fractional crystallization of a primary magma at depths of about 100 km.

INTRODUCTION

PETROLOGICAL STUDIES including microprobe analysis, major element determination, and melting experiments at high pressures have been carried out on the Apollo 14 high-alumina basaltic rock 14310 (139 and 22). On the basis of these studies, a possible origin of the lunar high-alumina basalt is discussed. Another Apollo 14 crystalline rock (14053,6) and a breccia (14321,22) have been studied petrographically, and some of their constituent minerals have been analyzed with the microprobe.

DESCRIPTION OF ROCKS AND MINERALS

Crystalline rock 14310

The rock is basaltic and shows intersertal texture, although some pyroxenes show ophitic or subophitic texture and only minor amounts of glass are present. The most abundant minerals are plagioclase, orthopyroxene, and strongly zoned clinopyroxene. Minor constituents are K-feldspar containing Ba, ilmenite, ulvö-

*Present address: Department of Geology, University of Otago, Dunedin, New Zealand.

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spinel, metallic iron, troilite, phosphate mineral (apatite or whitlockite), and Zr-rich minerals. The plagioclase consists of prismatic or needle-like crystals, which range in length from 0.1 to 1.0 mm. Some pyroxene crystals attain lengths of up to 2 mm. However, the grain size distribution appears to be continuous and a distinct phenocrystic texture is absent in the thin section examined. The mesostasis consists of very fine-grained crystals and a glass enriched in Si and K. The thin section contains one small patch (~1 mm across) of "cognate inclusion" (LSPET, 1971) consisting mainly of fine-grained plagioclase and pyroxene. The grain size of these minerals is one-fifth to one-tenth that of the main part. The boundary between the fine-grained part and the main part is sharp.

The abundances of major elements in rock 14310,139, obtained by the conventional wet-chemical analysis method, are given in Table 1. The rock is slightly undersaturated with silica, having 3.3 wt% normative olivine. The analysis shows high alumina and lime and relatively low iron. Alkalis, especially K_2O, and the MgO/FeO ratio are also higher than those of the Apollo 11 and 12 basaltic crystalline rocks. The composition is similar to those of some terrestrial high-alumina basalts, although Na_2O is still considerably lower and the alkali/alumina ratio of rock 14310 is much lower than those of the terrestrial high-alumina basalts (e.g., Kuno, 1960). It should be noted that the MgO/FeO ratios of the Apollo 15 gabbroic anorthosite 15415 (LSPET, 1972) and feldspathic basalts (Reid *et al.*, 1972) are also higher than those of the Apollo 11 and 12 crystalline rocks.

Crystalline rock 14053

The rock is coarse-grained and gabbroic (most crystals range from 0.5 to 2.0 mm and some clinopyroxenes attain 5 mm in length); however, plagioclase and clinopyroxene commonly show ophitic texture, and the rock may be called coarse-grained dolerite. The constituent minerals are plagioclase, strongly zoned clinopyroxene, olivine, pyroxferroite, K-feldspar, cristobalite, chromite, ulvöspinel, ilmenite, troilite, metallic iron, and phosphate mineral (apatite or whitlockite). The rock also contains mesostasis with Si- and K-rich glass in the interstices. In some of them a myrmekitic intergrowth of metallic iron, silica mineral (probably cristobalite), phosphate mineral, and Si- and K-rich glass is observed. The abundances of major elements in rock 14053 (LSPET, 1971) are significantly different from those in rock 14310 and are closer to those in the Apollo 12 crystalline rocks of intermediate MgO/FeO ratio, although iron and Ti are lower and Al and Ca are higher in rock 14053.

Table 1. Major elements and CIPW norm of rock 14310,139.

SiO_2, 46.88; TiO_2, 1.19; Al_2O_3, 21.68; FeO, 8.22; MgO, 7.42; CaO, 12.55; Na_2O, 0.72; K_2O, 0.50; Cr_2O_3, 0.25; MnO, 0.13; P_2O_5, 0.17; Total, 99.71 (wt%)
CIPW Norm
Or, 2.96; Ab, 6.10; An, 54.61; Di (Wo, 2.80; En, 1.57; Fs, 1.11); Hy (En, 14.34; Fs, 10.21); Ol (Fo, 1.83; Fa, 1.43); Il, 2.27; Cm, 0.38; Ap, 0.39 (wt%)

Analyzed by H. Haramura by the conventional wet-chemical analysis method.

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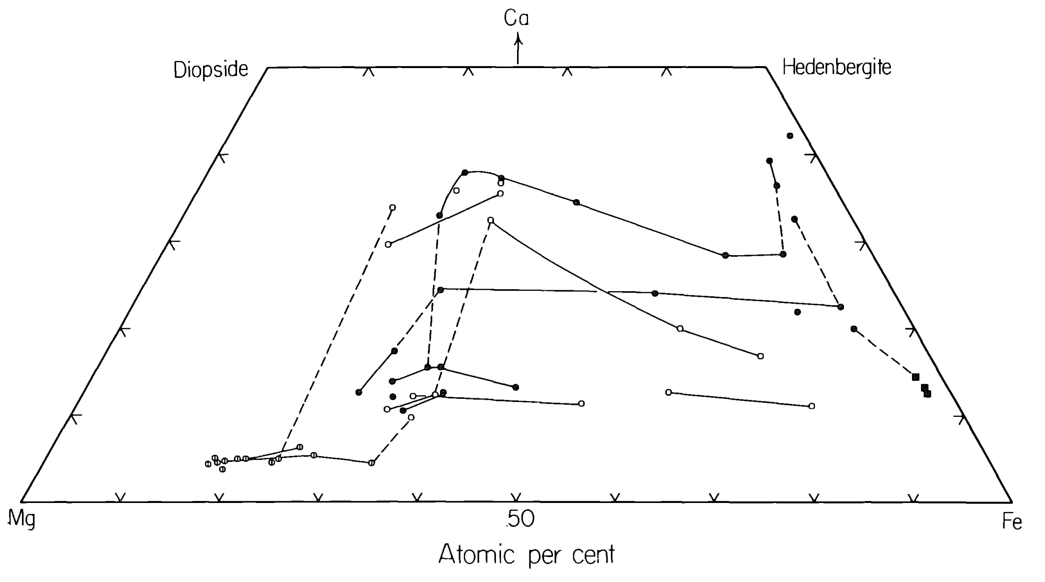

Fig. 1. Ca-Mg-Fe plot of the analyzed orthopyroxene, clinopyroxene, and pyroxferroite in rocks 14310,22 and 14053,6. Open and solid circles are clinopyroxenes in rocks 14310 and 14053, respectively; open circle with a vertical line and solid square are orthopyroxene (14310) and pyroxferroite (14053), respectively. Solid lines indicate the range of compositional zoning within single crystals, and dashed lines indicate discontinuous zoning (in the iron-rich side) and core-jacket relations.

Pyroxenes

Pyroxenes in rock 14310,22 are orthopyroxene, pigeonite, ferropigeonite, augite, and subcalcic ferroaugite. Their compositions are shown in the Ca-Mg-Fe diagram (Fig. 1), and selected microprobe analyses are given in Table 2. The method of microprobe analysis is the same as that described previously (Nakamura and Kushiro, 1970; Kushiro and Nakamura, 1970). Orthopyroxene (bronzite and hypersthene) occurs mostly as cores within pigeonite grains and is rarely in contact with augite. The Al content appears to decrease with increase of Fe/Mg ratio. Exsolution lamellae are not observed under the microscope in the orthopyroxene crystals. Pigeonite and ferropigeonite occur mostly as rims to the orthopyroxene. The Fe/(Mg+Fe) ratio ranges from 0.35 to 0.85, whereas the Ca content remains nearly constant (Wo, 10–12 mole %). Some pigeonite grains are zoned continuously to ferropigeonite; others are zoned discontinuously to (or surrounded by) augite, which is zoned continuously to subcalcic ferroaugite or ferropigeonite (Fig. 1). This may be due to the difference in the compositional zoning in different sectors of the pyroxene crystals, as shown by Hollister *et al.* (1971) and Boyd and Smith (1971) on the Apollo 12 pyroxenes. Augite occurs as rims to pigeonite or rarely to hypersthene. It is zoned to subcalcic ferroaugite near the margins. Some augite occurs as inclusions in pigeonite. The tie lines that connect the contacting Ca-poor pyroxenes and augite with core-jacket relations are shown in Fig. 1 (dashed lines). Although the analyses were made on the contacting pyroxenes within a range of 10 μ it is not certain that they are equilibrium tie lines.

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Table 2. Selected microprobe analyses of pyroxene, olivine, pyroxferroite, feldspar, and glass in rocks 14310,22, 14053,6, and 14321,22.

	14310,22										
	Orthopyroxene		Pigeonite and ferropigeonite		Augite		Plagioclase		K-feldspar	Glass	
	1	2	3	4	5	6	7	8	9	10	11
SiO_2	52.2	51.8	50.8	47.1	49.6	48.4	44.0	50.9	62.0	76.8	30.5
Al_2O_3	3.44	1.58	1.35	0.95	2.81	1.91	34.8	30.0	18.9	11.1	3.08
TiO_2	0.69	0.65	0.89	0.74	1.53	1.85	—	—	—	0.89	14.6
Cr_2O_3	0.61	0.48	0.39	0.08	0.72	0.04	—	—	—	—	—
FeO	11.2	16.1	19.8	34.1	13.5	18.3	0.08	0.39	0.25	0.86	35.7
MnO	0.21	0.28	0.33	0.47	0.26	0.29	—	—	—	0.04	0.36
MgO	28.4	24.8	20.3	9.17	16.4	11.4	—	—	—	0.02	3.16
CaO	2.50	3.11	5.07	5.74	14.0	16.4	19.1	14.5	0.37	0.73	7.09
Na_2O	0.02	0.03	0.02	0.04	0.09	0.11	0.64	2.86	0.90	0.94	0.40
K_2O	—	—	—	—	—	—	0.05	0.73	14.2	7.39	0.52
TOTAL	99.3	98.8	99.0	98.4	98.7	98.7	98.6	99.4	96.7*	98.8	95.4
Ca	4.9	6.2	10.4	12.7	29.5	35.3	94.1	70.5	1.9		
Mg	77.9	68.8	57.8	28.3	48.3	34.1	5.7(Na)	25.2(Na)	8.6(Na)		
Fe	17.2	25.0	31.8	59.0	22.2	30.6	0.2(K)	4.2(K)	89.4(K)		

	14053,6											14321,22	
	Pigeonite		Subcalcic augite and subcalcic ferroaugite				Hedenbergitic clinopyroxene	Pyroxferroite	Olivine	Plagioclase	Glass	Olivine	Orthopyroxene
	12	13	14	15	16	17	18	19	20	21	22	23	24
SiO_2	51.9	50.3	49.7	46.0	50.2	47.0	44.6	45.0	30.1	45.1	76.3	40.0	54.5
Al_2O_3	1.80	1.83	1.82	0.87	3.01	2.08	1.52	0.37	0.02	33.9	9.95	0.01	0.85
TiO_2	0.84	0.60	1.39	0.96	1.62	1.00	1.78	0.55	0.10	—	0.54	0.05	0.39
Cr_2O_3	0.82	—	0.60	0.08	—	—	0.03	—	0.01	—	—	—	—
FeO	17.3	25.7	18.3	39.0	15.4	34.7	30.2	45.0	63.4	0.29	1.83	10.7	13.7
MnO	0.38	0.41	0.37	0.65	0.29	0.53	0.49	0.68	0.79	—	—	0.11	0.21
MgO	21.3	14.7	15.9	1.90	14.2	2.80	0.53	0.80	4.97	—	—	50.1	29.5
CaO	6.43	6.20	11.7	9.87	15.4	12.5	18.0	6.39	0.34	18.8	0.85	0.06	0.94
Na_2O	0.02	0.00	0.09	0.03	0.00	0.00	0.12	0.06	0.12	0.88	0.30	—	—
K_2O	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.03	8.40	—	—
TOTAL	100.8	99.7	99.9	99.3	100.1	100.6	97.2	98.9	99.9	99.0	98.2	101.0	100.1
Ca	12.9	13.2	24.3	22.7	32.6	28.4	42.1	14.8	—	92.0		—	1.8
Mg	59.4	43.5	45.6	6.1	41.6	8.9	1.7	2.6	12.1	7.8(Na)		89.2	77.7
Fe	27.7	43.3	30.1	71.2	25.9	62.7	56.2	82.6	87.9	0.2(K)		10.8	20.5

*Low total is due to presence of about 3% BaO.

Pyroxenes in rock 14053 are pigeonite, augite, subcalcic augite, subcalcic ferroaugite, and hedenbergitic clinopyroxene. Their compositions are also shown in Fig. 1. Very narrow compositional gaps are observed between contacting pigeonite and subcalcic augite; the gaps are different for different pairs, however, and none of them appears to represent the equilibrium solvus. The augite and subcalcic augite are strongly zoned to very iron-rich subcalcic ferroaugite, which is discontinuously zoned to ferroaugite or hedenbergitic clinopyroxene. The very iron-rich subcalcic ferroaugite is commonly in contact with pyroxferroite (Fig. 1). The two strongly zoned clinopyroxene crystals in Fig. 1 show a strong iron enrichment; however, their Ca contents are significantly different, indicating metastable crystallization of

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pyroxenes from residual liquids that were isolated from each other and had different compositions during crystallization.

Breccia 14321,22 contains relatively large ($\sim$0.5 mm) fragments of homogeneous orthopyroxene with a much lower Ca content (Table 2, No. 24) than those of 14310,22. In the same thin section a pyroxene grain consisting of very thin lamellae (2–3 μ wide) of pigeonite and hypersthene is found. Their compositions are $Ca_{11}Mg_{56}Fe_{33}$ and $Ca_5Mg_{57}Fe_{38}$, respectively. The bulk composition of this pyroxene, obtained by using a defocused electron beam (45 μm across) and by moving the sample at a speed of 50 μm/min, is $Ca_{8.2}Mg_{56.9}Fe_{34.9}$, which is the composition of a low-Ca pigeonite. This evidence indicates that pigeonite (or high-clinopyroxene with $C_{2/c}$ symmetry) exsolved hypersthene with lowering temperature, supporting the shape of the pigeonite field suggested in the join enstatite-diopside (Kushiro, 1969, 1972). Similar exsolution has been found in orthopyroxene grains from rock 14310 (Finger *et al.*, 1972).

The Al-Ti relations of the analyzed pyroxenes from rocks 14310 and 14053 are shown in Fig. 2. The Ti content in orthopyroxene from 14310 is relatively low and uniform (between 0.02 and 0.01 on the basis of 6 oxygens) and appears to be independent of Al content. Pigeonite in both 14310 and 14053 also has a relatively low Ti content (mostly below 0.03). The Ti contents of these analyzed pyroxenes and also those from the Apollo 12 crystalline rocks appear to increase systematically with an increase of the Ca/(Ca+Mg+Fe) ratio (or wollastonite component) up to a value of 0.32 (Fig. 3) and are independent of the Fe/Mg ratio.

In the zoned clinopyroxenes, the Ti/Al ratio increases with increase of the Fe/Mg

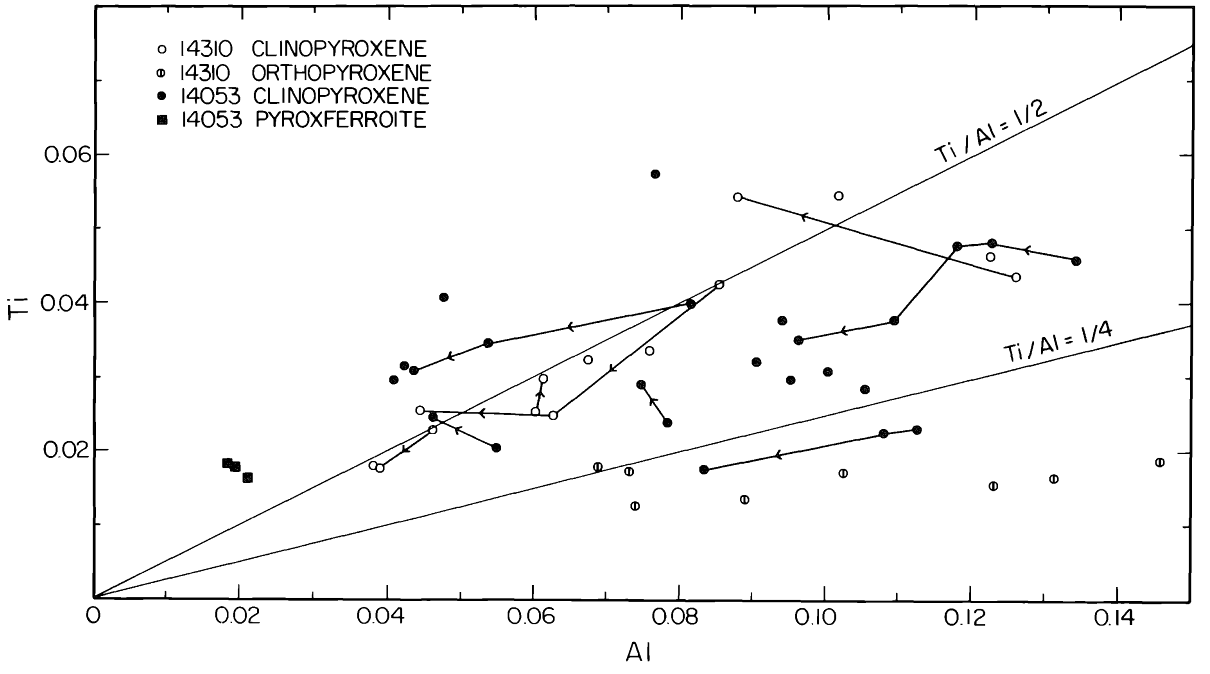

Fig. 2. Al-Ti relations of the analyzed clinopyroxenes in rocks 14310 and 14053. Symbols as in Fig. 1. Solid lines indicate the variations of Al and Ti in the zoned clinopyroxenes. Values based on 6 oxygens.

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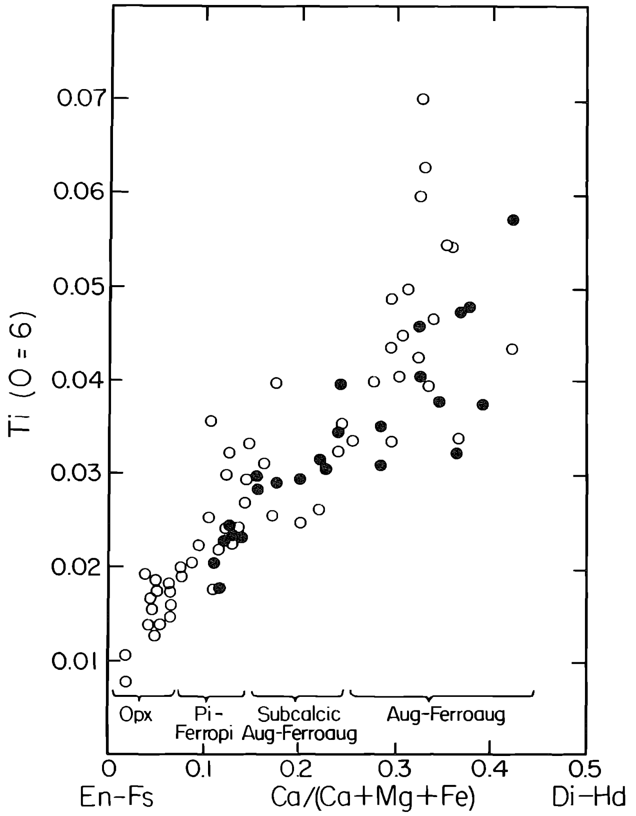

Fig. 3. The relations between Ti content and Ca/(Ca + Mg + Fe) ratio in the analyzed pyroxenes in rocks 14310, 14053, and 14321 (open circles) and those in the Apollo 12 crystalline rocks 12018, 12020, 12064, and 12065 (solid circles) (Kushiro and Nakamura, 1970).

ratio, and in very iron-rich clinopyroxenes and pyroxferroite this ratio is larger than 1/2 (Fig. 2). The Ti/(Al+Cr) ratio is also larger than 1/2 in most of these iron-rich clinopyroxenes and pyroxferroite. If all Ti is Ti^{4+}, the charge in the octahedral site is in excess of that in the tetrahedral site; that is, the charge balance is not maintained because it is unlikely that Fe^{3+} exists in these pyroxenes and pyroxferroite or that Ti^{4+} enters into the tetrahedral site. It seems most likely that a part of Ti is Ti^{3+}, at least in these very iron-rich pyroxenes and pyroxferroite. This possibility has been suggested previously by Boyd and Smith (1971) and Bence and Papike (1972).

Feldspar

Plagioclase is the most abundant mineral in rock 14310. Its composition ranges from $An_{94.1}Ab_{5.7}Or_{0.1}$ to $An_{70.5}Ab_{25.2}Or_{4.2}$. Selected analyses of plagioclase are given in Table 2. Plagioclase generally shows normal zoning, and in one case a single plagioclase crystal covers the above compositional range. The K content increases regularly with increasing Na content. Brown and Peckett (1971) distinguished phenocryst and groundmass plagioclase in two thin sections of 14310 and have shown that the phenocryst plagioclase is zoned to more sodic and potassic compositions, whereas the groundmass laths are as low in sodium and potassium as are the phenocryst cores. However, there are some exceptions; in the present analyses some of the very small plagioclase laths ("groundmass plagioclase") have composi-

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tions $An_{74}Ab_{26}$ and $An_{84}Ab_{16}$, which are considerably more sodic than those reported by Brown and Peckett (1971). Plagioclase in rock 14310 is often in contact with K-feldspar, which shows a compositional zoning, $An_{5.7}Ab_{12.8}Or_{81.5}$ to $An_{3.1}Ab_{6.4}Or_{90.5}$, and contains 3 to 6 wt% BaO. The K content of bytownite in contact with K-feldspar, but outside the effect of K emission from K-feldspar, is relatively high compared with that of bytownite of the same Na/Ca ratio in terrestrial basaltic rocks. This may be due to the fact that in the terrestrial igneous rocks more sodic plagioclase is in equilibrium with K-feldspar and bytownite is usually not saturated with K. Plagioclase in the "cognate inclusion" of rock 14310,22 has the composition $An_{92.3}Ab_{7.1}Or_{0.6}$, which is nearly the same as that of the calcic core of the zoned plagioclase in the main part.

Plagioclase in rock 14053,6 is relatively homogeneous ($An_{91}Ab_9$–$An_{93}Ab_7$) except near the rim, which is zoned to sodic bytownite and is commonly in contact with K-feldspar, cristobalite, and glass. The compositions of one pair of contacting plagioclase and K-feldspar are $An_{70.6}Ab_{24.9}Or_{4.5}$ and $An_{1.0}Ab_{1.8}Or_{97.2}$, respectively. The K-feldspar is slightly zoned from $An_{1.9}Ab_{2.0}Or_{96.0}$ in the core to $An_{1.0}Ab_{1.8}Or_{97.2}$ at the rim.

Plagioclase in an anorthositic rock fragment in breccia 14321,22 shows a compositional zoning from $An_{90.8}Ab_{8.6}Or_{0.6}$ in the core to $An_{82.4}Ab_{15.6}Or_{2.0}$ at the rim.

Olivine

Rock 14053,6 contains olivines of two distinctly different compositions. Inclusions of olivine in pigeonite crystals have the composition $Fo_{63}Fa_{37}$, whereas olivine that occurs in the interstices and is associated with cristobalite has the composition $Fo_{12}Fa_{88}$. Olivines of intermediate compositions have not been found, suggesting a reaction relation between magnesian olivine and pigeonite, and breakdown of the ferrosilite component in iron-rich pyroxene to iron-rich olivine and silica.

In breccia 14321,22 two fragments of zoned olivine crystals have cores with the compositions $Fo_{91}Fa_9$ and $Fo_{89}Fa_{11}$. They are more magnesian than those of the Apollo 11 and 12 crystalline rocks and nearly the same as the magnesian olivines crystallized from some terrestrial basic magmas, suggesting that at least some lunar basic magmas had MgO/FeO ratios similar to those of terrestrial basic magmas.

Glass

Pale brown to colorless glass is commonly found in the mesostasis of rock 14310,22. The composition determined with the microprobe is enriched in Si and K and is similar to those of the mesostasis of the Apollo 11 crystalline rocks (e.g., Roedder and Weiblen, 1970; Kushiro and Nakamura, 1970). In some of the mesostasis, colorless glass spherules are surrounded by a dark part, which consists of fibrous or skeletal ilmenite and very fine-grained pyroxenes. These appear to be quench crystals; that is, the dark mafic part was originally liquid. It seems likely that the colorless glass and the dark part were formed by liquid immiscibility, as first reported by Roedder and Weiblen (1970) in the Apollo 11 crystalline rocks.

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The composition of the colorless glass (Table 2, No. 10) also is enriched in Si and K. The bulk composition of the quenched mafic part (Table 2, No. 11) is very rich in Fe and Ti, and its norm indicates that it is essentially a mixture of ilmenite and pyroxene. The ratio of the mafic part to the colorless glass is not large on the whole, and the bulk composition of the residual liquid is closer to that of the Si- and K-rich glass. These residual liquids would have been produced by small-scale fractionation of the magma, as suggested by Kushiro *et al.* (1971). However, the glass of such composition has considerably higher Or/Ab ratios than most terrestrial granites, and its composition is similar to those of some alaskite and aplite.

In rock 14053, glass patches also are found in the interstices and are commonly associated with pyroxferroite. Their compositions are very similar to those of rock 14310; i.e., of potassic granite or rhyolite (Table 2, No. 22).

MELTING EXPERIMENTS

Melting experiments have been carried out on high-alumina basalt 14310 at pressures of 5 to 25 kb and temperatures of 1100° to 1470°C in the piston-cylinder apparatus with graphite capsules. The piston-out technique, without correction, was applied to all the runs. The error of pressure in the low-pressure range (< 10 kb) should be very large (probably up to 20%). Synthetic glass of this rock composition made from oxides (Table 1) was used as a starting material for preliminary determination of the phase relations. The results of the runs are shown in Table 3 and in a pressure-temperature diagram (Fig. 4). The results on both the natural rock and the synthetic glass are consistent with each other, except that plagioclase is finer

Table 3. Results of experiments on rock 14310,139 and a synthetic glass of the same composition.

Pressure (kb)	Temperature (°C)	Time (hr)	Results
			14310,139 (homogenized fines)
5	1300	1	Pl + Gl
5	1250	2	Pl + Gl
5	1200	$3\frac{1}{2}$	Pl + Px (Opx + Cpx?) + Sp + Gl (Opx: 4.3 wt% Al_2O_3)
7.5	1350	$\frac{1}{2}$	Gl + q-Cryst
7.5	1315	1	Gl
7.5	1290	$1\frac{1}{2}$	Pl + Gl
7.5	1250	2	Pl + Px (Opx + Cpx?) + Gl
10	1350	1	Gl
10	1325	1	Gl
10	1300	$1\frac{2}{3}$	Pl + Sp (rare) + Gl
10	1275	2	Pl + Sp + Px (trace) + Gl (Sp: 20.0 wt% Cr_2O_3, 0.4% TiO_2, and 17.5% MgO)
10	1250	2	Px (Opx + Cpx?) + Pl + Sp (rare) + Gl (Opx: Ca 5.1, Mg 78.5, Fe 16.4, with 6.6 wt% Al_2O_3)
10	1175	2	Pl + Px (Opx + Cpx) + Sp + Gl (trace)
15	1325	1	Sp + Gl
17.5	1300	$1\frac{1}{2}$	Gar + Cpx + Pl + Sp (trace) + Gl
20	1375	1	Gl (some Gl fragments contain very rare Sp)
20	1325	$1\frac{1}{2}$	Gar + Cpx + Pl + Sp (trace) + Gl (Cpx: Ca 40.5, Mg 47.1, Fe 12.4, with ~11% Al_2O_3; Gar: Ca 20.6 Mg 57.9, Fe 21.6)
25	1390	$\frac{2}{3}$	Gar + Cpx (rare) + Gl

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Table 3. (continued).

Pressure (kb)	Temperature (°C)	Time (hr)	Results
			14310 synthetic glass
5	1325	1	Gl
5	1225	2	Pl + Px + Sp + Gl
5	1125	4	Pl + Px + Sp
5	1100	$4\frac{1}{6}$	Pl + Px + Sp
7.5	1360	$\frac{2}{3}$	Gl
7.5	1300	$1\frac{1}{2}$	Pl + Sp + Gl
7.5	1200	$2\frac{1}{2}$	Pl + Px + Sp + Gl
12.5	1325	1	Sp + Gl
12.5	1300	$1\frac{1}{2}$	Pl + Sp + Px? + Gl
15*	1350	$1\frac{1}{2}$	Sp (trace) + Gl
15	1325	$1\frac{1}{2}$	Sp + Gl
15	1300	$1\frac{2}{3}$	Pl + Cpx + Sp + Gl
15	1225	$2\frac{1}{5}$	Pl + Px + Gar + Sp
15	1175	$3\frac{1}{3}$	Pl + Px + Gar
17.5	1325	$1\frac{5}{6}$	Sp + Cpx + Gl
20*	1350	$1\frac{1}{2}$	Gar + Cpx + Sp (rare) + Gl
20	1250	2	Gar + Cpx + Sp + Gl
20	1200	3	Pl + Cpx + Gar + Sp
30†	1465	$\frac{1}{2}$	Gar (rare) + Gl
30†	1425	1	Gar + Gl

Abbreviations: Cpx, clinopyroxene; Gar, garnet; Gl, glass; L, liquid; Ol, olivine; Opx, orthopyroxene; Pl, plagioclase; Px, pyroxene (Opx and Cpx are not distinguished unless specifically mentioned); q, quench; Sp, spinel.

*Conducted in iron capsule. Others conducted in graphite capsule.

†Conducted by S. Akimoto in graphite capsule with tetrahedral anvil apparatus.

grained and the amount of spinel (a solid solution of spinel, hercynite and chromite) appears to be larger in runs made with the synthetic glass than in those made using the natural rock.

Below about 10 kb, calcic plagioclase is the liquidus phase and is followed by chromian spinel, aluminous orthopyroxene, and clinopyroxene with lowering temperature. Between about 10 and 20 kb, spinel is the liquidus phase, and above about 20 kb, pyrope-rich garnet is the liquidus phase. In these experiments spinel crystallizes over a wide P-T range. It is almost colorless to pale brown near the liquidus, becoming brown to dark brown with lowering temperature to near the solidus and dark brown to almost opaque below the solidus. Clearly its composition changes with temperature. Some investigators (e.g., Ford *et al.*, 1972; Walker *et al.*, 1972) reported that spinel crystallizes on the liquidus but in a much narrower pressure range (e.g., 12–17 kb). This discrepancy may be due to a difference in oxygen fugacity during the run. In the present experiments, made in graphite capsules, oxygen fugacity would have been higher than in those made with molybdenum and iron capsules (Heubner, 1971), and spinel would therefore be stable over a wider P-T range. The melting interval of rock 14310 is 150°–200°C in the pressure range studied.

Several runs were made at 1 atm on the synthetic glass in a CO_2 atmosphere by T. Katsura. Under these conditions, olivine crystallizes between 1213° and 1199°C. In the runs made at lower oxygen fugacity olivine also crystallizes near 1200°C (Muan *et al.*, 1972; Ford *et al.*, 1972; Ringwood *et al.*, 1972). Since olivine is not

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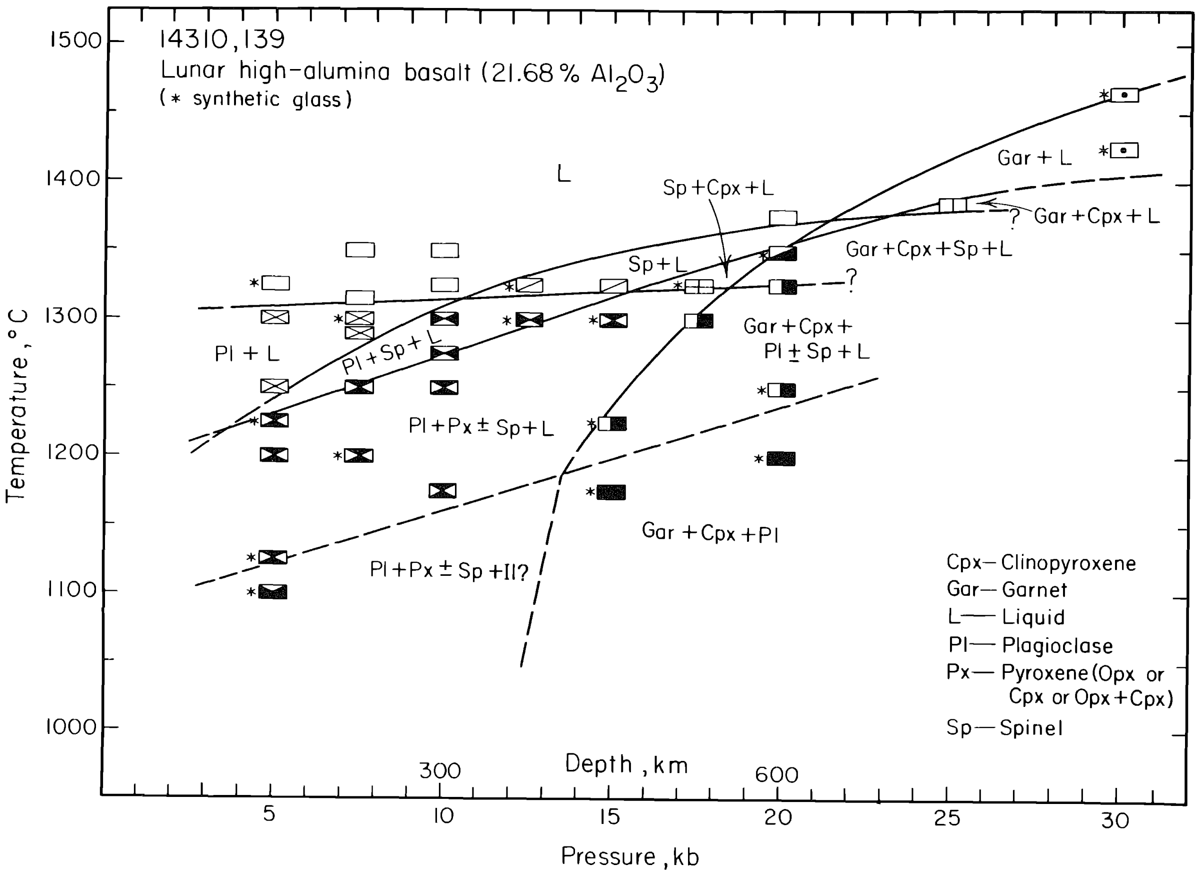

Fig. 4. Results of high-pressure experiments on rock 14310,139 and a synthetic glass of the same composition. The runs made with the latter starting material are shown by asterisks. Different box symbols correspond to different assemblages that are labeled in each field of the diagram.

observed at pressures higher than 5 kb, the P-T field for olivine crystallization terminates below 5 kb, as shown by Ford *et al.* (1972).

Orthopyroxene crystallized at 10 kb and 1250°C has the composition $Ca_{5.1}Mg_{78.5}Fe_{16.4}$ with 5.7 to 7.7 (average 6.6) wt% Al_2O_3 and 0.82 wt% Cr_2O_3. These amounts are greater than those of the orthopyroxene in rock 14310. At 5 kb and 1200°C the orthopyroxene contains 4.3 wt% Al_2O_3, which is still greater than that of the 14310 orthopyroxene. Spinel crystallized at 10 kb and 1275°C contains about 20.0 wt% Cr_2O_3, 48.5% Al_2O_3, 17.5% MgO, 13.5% FeO, and 0.4% TiO_2. Clinopyroxene crystallized at 20 kb and 1325°C has the composition $Ca_{40.5}Mg_{47.1}Fe_{12.4}$ with high alumina contents (~11% Al_2O_3 and 0.35% Na_2O). The Ca:Mg:Fe ratio of this clinopyroxene is almost identical with that of a core augite (~$Ca_{40}Mg_{47}Fs_{13}$) found in orthopyroxene in rock 14310 by Hollister *et al.* (1972), although alumina is much higher and Ti is lower in the former than in the latter.

DISCUSSION

The experimental results indicate that the 14310 composition is within one of the volumes of alumina-rich phases in a pressure range from 1 atm to at least 30 kb;

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that is, in the plagioclase volume at low pressures, in the spinel volume at intermediate pressures, and in the garnet volume at high pressures. At about 10 kb this composition lies on the boundaries between the plagioclase and spinel volumes, and at about 20 kb it lies between the spinel and garnet volumes. These relations are shown in the simple system anorthite-olivine-silica (Fig. 5), onto which the 14310 composition is projected from the apex of other components (normative Or, Ab, Wo, Il, Cm, and Ap). The total of these remaining components is about 15 wt% of the 14310 composition. At 1 atm the 14310 composition is in the plagioclase field, whereas at 10 kb it lies on or very close to the spinel-anorthite boundary. The phase boundaries at 10 kb are consistent with the runs made on 14310 at this pressure. The compositions of feldspathic basalts (or anorthositic gabbro), which Reid *et al.* (1972) have suggested to be lunar highland materials, also are plotted in this system. The compositions of rock 14310 and feldspathic basalts are on the An side of the piercing points at 1 atm, Sp + Ol + An + *L* (*C*) and Ol + An + Ca-poor Px + *L* (*D*), and those at 10 kb, Ol + Sp + Opx + *L* (*A*) and Sp + Opx + An + *L* (*B*), or the low-temperature cotectic boundaries at least to 10 kb. Walker *et al.* (1972) have indicated that the composition of the fines sample 14259 is close to the three-phase (olivine, pyroxene, and plagioclase) saturation boundary (or piercing point) at 1 atm. Addition of 20 wt% plagioclase ($An_{94}Ab_6$) to the fines gives a composition almost identical with that of rock 14310, indicating that the composition of the latter is away from the composition of the piercing point by about 20 wt% calcic plagioclase component. On the other hand, the possible lunar "mantle" materials (peridotite or pyroxenite with or without a small amount of plagioclase) would lie on the olivine or pyroxene side of these piercing points. If rock 14310 is not a plagioclase-cumulate rock, it cannot be a product of direct partial melting of the lunar "mantle" materials at least to 300 km under anhydrous conditions; nor is it a product of fractional crystallization of magma ("primary magma") that was formed by direct partial melting of the lunar "mantle" materials. Even if rock 14310 is a plagioclase-cumulate rock, these arguments would still be valid unless cumulate plagioclase is more than 15 (or close to 20) wt%, because the composition 14310, with less than 15 wt%

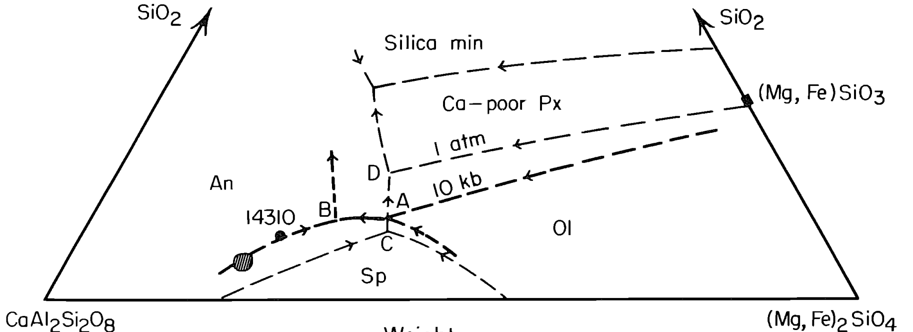

Fig. 5. System $(Mg,Fe)_2SiO_4$-$CaAl_2Si_2O_8$-SiO_2(Fe/[Mg + Fe] ratio, 0.3) at 1 atm and 10 kb. The liquidus boundaries are based on Andersen (1915), Roeder and Osborn (1966), and present experimental results. Solid circle, 14310 composition; dashed area, feldspathic basalts (Reid *et al.*, 1972). A and B are piercing points at 10 kb; C and D are those at 1 atm.

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anorthite, would still be on the An side of these piercing points at pressures at least up to 20 kb.

Therefore, a two-stage model must be considered for the origin of rock 14310 and the feldspathic basalts. One possibility is that the magmas of these rocks were formed by bulk melting or partial melting of a plagioclase-cumulate rock, which had been formed in an earlier stage by a large-scale differentiation of magma in the shallow parts of the lunar interior and constituted a part of the lunar highland.

If rock 14310 is a plagioclase-cumulate rock with plagioclase enrichment by more than 15 (or close to 20) wt%, the rock could be formed either by partial melting of the lunar "mantle" material or by fractional crystallization of a more mafic magma, followed by plagioclase cumulation before the solidification of the magma. This possibility has been discussed by Walker *et al.* (1972). As described before, however, such a large amount of plagioclase phenocrysts was not observed in the thin section examined (14310,22). Brown and Peckett (1971) estimated about 10 volume % of plagioclase phenocrysts in two thin sections of rock 14310, although definition and estimation of plagioclase phenocrysts is extremely difficult in this rock, because the grain size distribution appears to be continuous. Because the discrepancy between the amount of plagioclase phenocrysts suggested and that required for the plagioclase-cumulation hypothesis appears to be significantly large, and because resorption of cumulate plagioclase was unlikely to occur during the ascent of magma, it is difficult to accept the plagioclase-cumulation hypothesis. The process involving garnet (at depths greater than 500 km) may not be related to the origin of this high-alumina basalt.

The above discussion is based on the assumption that there was no loss or gain of alkalis and volatile components during the magmatic stage. Brown and Peckett (1971) observed that "groundmass" plagioclase is more calcic than the outer part of "phenocryst" plagioclase, and to explain this observation they suggested significant loss of alkalis from the magma at or near the lunar surface. They estimated the loss of Na_2O and K_2O to be about 70 wt% of the amount originally present. If this is true, the discussion on the origin of rock 14310 may be subject to change. To examine

Table 4. Results of experiments on 14310 composition with alkalis added (3.2 wt% Na_2O and ~1% K_2O)

Pressure (kb)	Temperature (°C)	Time (hr)	Results
2.5	1215	$2\frac{1}{3}$	Pl + Sp (rare) + Ol (rare) + Gl
3.5	1220	$2\frac{1}{3}$	Ol (rare) + Sp (rare) + Gl (Ol, ~$Fo_{90}Fa_{10}$)
3.5*	1215	$3\frac{1}{2}$	Pl + Sp + Ol? + Gl
3.5*	1200	2	Pl + Sp + Px (mostly Opx) + Ol? + Gl
5	1240	$1\frac{5}{6}$	Sp + Gl
5	1225	$1\frac{1}{2}$	Sp + Pl (rare) + Gl
5*	1210	$2\frac{1}{3}$	Pl + Sp + Px + Ol + Gl
10	1300	$\frac{5}{6}$	Sp + Gl
10*	1275	2	Sp + Gl
10*	1250	1	Sp + Px + Gl
10*	1225	2	Sp + Pl + Px + Gl

*Synthetic glass with alkalis added. Others are 14310,139 with alkalis added.

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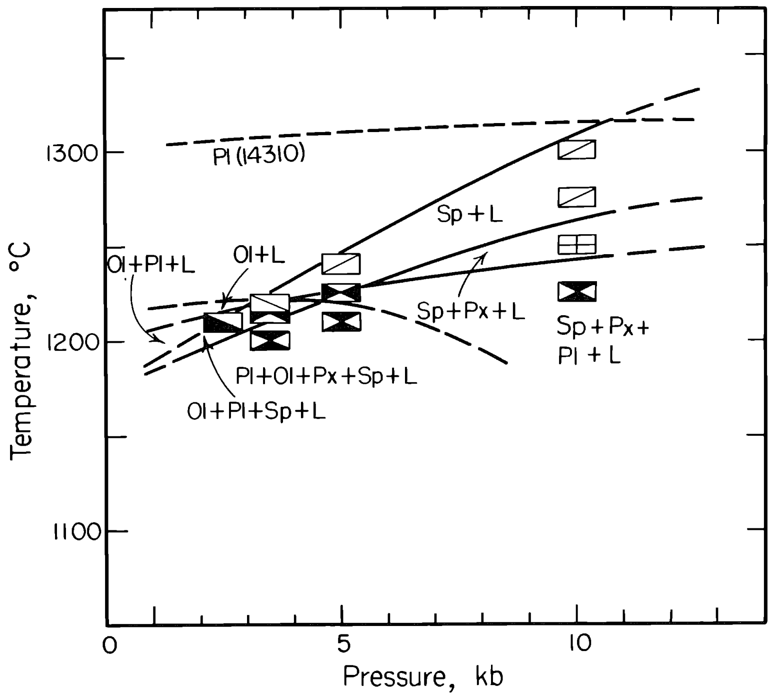

Fig. 6. Results of experiments on the alkali-enriched 14310 composition (total alkalis, Na_2O 3.2 and K_2O ~1.0 wt%). Dashed curve indicates the liquidus of plagioclase for the 14310 composition in Fig. 4. Symbols as in Fig.4 except presence of olivine in the low-pressure runs.

this possibility, melting experiments have been carried out on synthetic glass of the 14310 rock composition and on actual rock 14310,139 with addition of 2.5 wt% Na_2O and about 0.5% K_2O, making total alkali contents 3.2 wt% Na_2O and about 1% K_2O. These contents are similar to those of some terrestrial high-alumina basalts. The results are given in Table 4 and Fig. 6. The major changes are a drastic drop of the plagioclase liquidus (by 70°–90° between 2.5 and 10 kb) and crystallization of olivine to higher pressures (at least up to 5 kb). At 3.5 kb the liquidus temperature is slightly above 1220°C and four crystalline phases—olivine, spinel, plagioclase, and pyroxene—appear within a 20° interval, indicating that the alkali-enriched 14310 composition is very close to the boundary of the volumes of these four phases. This evidence suggests that the liquid of this alkali-enriched composition could be formed by direct partial melting of a plagioclase- and spinel-bearing peridotite at about 3 kb (~100 km) or by fractional crystallization of a more mafic magma. If significant alkali loss is accepted, a single-stage model may be possible for the origin of rock 14310; the original magma of rock 14310 was formed at depths near 100 km and erupted on the lunar surface, where alkalis were lost before the crystallization of most plagioclase and other minerals. The crystallization of plagioclase after alkali loss must have been very rapid, because the magma became supercooled with respect to plagioclase as soon as alkalis were lost. Most of the other crystals also would have crystallized rapidly.

The above discussion is based on the assumption that the lunar interior was completely or almost free of H_2O when the magmas were generated. If H_2O existed

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in the lunar interior, the magma would have been considerably enriched in An component, as shown by Yoder (1965), and the origin of An-rich magma could be explained by a single-stage process without considering a large amount of alkali loss. However, there is no positive evidence for the existence of a significant amount of water in the lunar interior.

As mentioned before and pointed out by other investigators (e.g., Ford *et al.*, 1972; Ringwood *et al.*, 1972), the MgO/FeO ratio of rock 14310 is greater than that of any of the mare-type crystalline rocks collected by the Apollo 11, 12, and 14 missions. If the magma of this rock were directly derived from the lunar "mantle," as in the single-stage model, the lunar "mantle" may have a MgO/FeO ratio similar to that of the earth's mantle. Olivine crystallized at 3.5 kb and 1220°C from the alkali-enriched 14310 composition is about $Fo_{90}Fa_{10}$, supporting this argument. Even for the double-stage model, the same conclusion would be obtained. As described before, the most magnesian parts of the zoned olivine fragments in breccia 14321 have the compositions $Fo_{91}Fa_9$ and $Fo_{89}Fa_{11}$. If they crystallized from magmas that were derived initially from the lunar interior, the lunar "mantle" should have a similar or higher MgO/FeO ratio, at least in the source regions of the high-alumina basalts of the lunar highlands.

In summary, if significant alkali loss did not occur before or during the solidification of magma of rock 14310, the rock could have been formed by remelting (either bulk melting or partial melting) of a plagioclase-cumulate rock that had been formed by a large-scale differentiation in the shallower part of the lunar interior in an earlier stage. Alternatively, if significant alkali loss ($\sim$70% or more) occurred, the rock could have been formed either by partial melting of the lunar "mantle" materials or by fractional crystallization of a more primitive magma at depths of about 100 km.

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